

Calibration Limits and Resource Costs in Sub THz Pollutant Sensing

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Abstract—We study how persistent spectral calibration error changes conditional pollutant retrieval in a layered HITRAN model with real Beijing concentration inputs and simulated radio attenuation. A convex linear estimator balances atmospheric bias, calibration sensitivity and noise at fixed resources. Exact rational dual certificates for the represented normalized model give a noise-free NO₂ error floor of 1.984 at an assumed persistent error of $\pm 10^{-4}$ dB, above the unit magnitude-normalized target regardless of pilot count. At ten million ideal simultaneous pilots, calibration-aware estimation improves the 64-state stress-replay NO₂ error from 3.385 to 2.425 but does not restore four-gas feasibility. CO and SO₂ retain useful conditional margins. A separate 71-tone sequential acquisition model charges dwell time and published test-port power explicitly. These results identify calibration and acquisition requirements; the error envelope, coherent receiver and field retrieval remain unvalidated.

Index Terms—Atmospheric sensing, calibration, generalized least squares, HITRAN, integrated sensing and communication.

I. QUESTION AND CONTRIBUTION

Shared communication and absorption sensing has a precedent for atmospheric water vapour [1]; multi-medium propagation models motivate treating molecular and particle attenuation jointly [2]. Ambient pollutant retrieval additionally requires separating a weak target spectrum from background uncertainty. A small local lower bound alone cannot establish an achievable measurement error.

We ask which error requirements follow when that missing estimator is supplied. Our contribution is a conditional joint atmospheric/calibration estimator, a machine-checkable represented-model bias lower bound, and an acquisition model that charges sequential dwell and published source power. The core result tests whether more pilots can repair the declared persistent-error envelope. Generalized least squares and Fisher information are established methods [14]; this work applies them to the declared spectroscopy and link rather than claiming a new estimation principle. Earlier probe-selection, satellite-column and regression studies are retained in the appendix as supporting diagnostics.

II. EVIDENCE AND FIXED PHYSICAL MODEL

The primary spectroscopy input is the archived 9,340-line HITRAN table, checked against HAPI using the same line records [5], [6]. Real UCI Beijing records provide measured surface concentrations and weather [8]. They do not provide vertical profiles or synchronized radio attenuation. The chronological split and concentration-order exclusions are unchanged. All current atmospheric summaries use training

rows only; both training quantile scenarios and reused test-period regressions are explicitly separate from independent field validation.

Four reported errors are normalized by fixed concentration magnitudes: CO 4,000, O₃ 100, SO₂ 40 and NO₂ 25 $\mu\text{g m}^{-3}$. These numbers originate in WHO guideline values [12]; they are magnitude references, not detection probabilities, instantaneous compliance thresholds or a common averaging-period specification. Simulated estimates are not treated as health measurements.

The 24-layer atmosphere extends from 0 to 12 km. Line intensities, partition sums, pressure and Doppler broadening enter a Voigt absorption calculation with physical number-density units. The source surface concentration is mapped to an exponential gas profile with 1.5 km scale height. Water uses a 2 km profile; PM uses 1 km and the previously declared exploratory optical parameters. These profiles are assumptions. Reference temperature, pressure and dew point are taken from the training-only protocol. The explicit equations, line exclusions and prior implementation checks appear in the appendix.

The reference link for the full-grid comparison uses 256 conceptual frequencies from 60 to 400 GHz, 550 km altitude, 45° elevation, 23 dBm total transmitted power, 0.5/0.3 m transmit/receive apertures, 1 MHz per probe, 6 dB noise figure and 5 dB implementation loss. The full-grid estimator retains 202 probes with reference SNR at least 5 dB; the sequential comparison below uses 71 of these in its single band. Their power is not redistributed after masking. Thus the current study does not silently improve the link when opaque probes are removed. The frequency grid is a multiband modeling device, not a demonstrated contiguous waveform.

III. ATTAINABLE ESTIMATION UNDER DECLARED NOISE

After subtracting the reference background, use

$$y = D\theta + N\beta + n, \quad n \sim \mathcal{N}(0, \Sigma). \quad (1)$$

Columns of D contain the four magnitude-normalized gas spectra. N contains an offset, reference background scale and both PM spectra. The covariance is fixed and known in this conditional model; SNR is evaluated at the reference background, not re-estimated from unknown target values. A parameter-dependent physical noise model would require a different likelihood and Fisher calculation.

For linear SNR γ_i , $c = (10/\ln 10)^2$ and N_p pilots, the two declared thermal terms are

$$\begin{aligned} v_i^{\text{power}} &= c(1 + \gamma_i^{-1})^2/N_p, \\ v_i^{\text{coherent}} &= 2c/(N_p\gamma_i). \end{aligned} \quad (2)$$

The power approximation is motivated by the pilot model in [7]; neither observation law has been calibrated for this receiver. Residual covariance is added as $\Sigma = \text{diag}(v) + \sigma_r^2 C$. We test $C = I$ and $C_{ij} = (1 - \rho)1\{i = j\} + \rho e^{-|f_i - f_j|/10 \text{ GHz}}$ with $\rho = 0.9$. Correlation direction and nuisance alignment can increase or decrease information. An independent residual standard deviation is not a systematic calibration specification.

Let $LL^T = \Sigma$, $\tilde{D} = L^{-1}D$, and let Q be an orthonormal basis for $L^{-1}N$. Set $P = I - QQ^T$ and

$$H = (\tilde{D}^T P \tilde{D})^{-1} \tilde{D}^T P L^{-1}. \quad (4)$$

If $P\tilde{D}$ has full target rank, $HD = I$, $HN = 0$ and

$$\hat{\theta} = Hy, \quad \mathbb{E}\hat{\theta} = \theta, \quad V = H\Sigma H^T = (\tilde{D}^T P \tilde{D})^{-1}. \quad (5)$$

These identities follow by substitution and $P^2 = P = P^T$. They attain the efficient Gaussian information bound in this unconstrained linear model [14]. They do not guarantee a nonnegative estimate or transfer to a misspecified nonlinear receiver. Our implementation uses Cholesky whitening and an SVD solve, not explicit normal-matrix inversion. Normalized nuisance and target rank tests use tolerance 10^{-12} ; a target absorbed by nuisance is rejected rather than assigned a finite precision.

IV. SYSTEMATIC CALIBRATION AND FORWARD-MODEL BIAS

If the actual mean differs by b , the same estimator obeys exactly

$$\text{bias}(\hat{\theta}) = Hb, \quad \text{MSE}_j = V_{jj} + (h_j^T b)^2, \quad (6)$$

where h_j^T is row j of H . When the fitted covariance is wrong, replace V by $H\Sigma_{\text{true}}H^T$. This sandwich calculation is included in the covariance experiment.

For an arbitrary persistent spectral error $\|b\|_\infty \leq \epsilon$, the exact targetwise worst case is

$$\sup_b \text{MSE}_j = V_{jj} + \epsilon^2 \|h_j\|_1^2. \quad (7)$$

The upper bound follows from the triangle inequality and is attained by $b_i = \epsilon \text{sign}(h_{ji})$. Different targets may attain their maxima at different bias vectors. For normalized RMSE at most one, the allowable deterministic error is therefore

$$\epsilon_j = \frac{\sqrt{1 - V_{jj}}}{\|h_j\|_1}, \quad V_{jj} \leq 1. \quad (8)$$

No allowance exists if thermal/residual noise already exceeds the target. An offset or modeled background-scale bias lies in $\text{col}(N)$ and is removed; this worst case concerns arbitrary bias shapes outside that nuisance model. Equations (6)–(8) are self-derived conditional sensitivity results, not measured calibration bounds.

V. JOINT ATMOSPHERIC AND PERSISTENT CALIBRATION ROBUSTNESS

Let B contain the 54 previously archived atmospheric discrepancy spectra. We now declare the mean-error set

$$\mathcal{B} = \text{conv}\{\pm b_s : 1 \leq s \leq 54\} + \{u : \|u\|_\infty \leq \epsilon\}.$$

The first component represents the finite atmospheric design envelope; the second is a persistent per-frequency calibration error, not independent noise that averages down with pilots. Neither component is an empirically calibrated confidence region. For a row h^T satisfying $h^T D = e_j^T$ and $h^T N = 0$, its exact worst squared error under diagonal variance v is

$$R_j^2(h) = \sum_i v_i h_i^2 + \left(\max_s |h^T b_s| + \epsilon \|h\|_1 \right)^2. \quad (9)$$

The support function of the symmetric finite hull is $\max_s |h^T b_s|$; the box support is $\epsilon \|h\|_1$. Their sum is attained independently, which proves (9). Worst cases for different gases need not be the same physical state. Minimizing this convex objective over the equality constraints gives the calibration-aware linear estimator. We retain the complete equality nullspace, because restricting it to the span of atmospheric errors can miss directions relevant to the ℓ_1 penalty. This is a specialization of convex robust estimation, not a new general optimization principle.

The grid uses $\epsilon = 0, 10^{-5}, 10^{-4}, 10^{-3}$ dB, independent random residual SD 0 or .001 dB, and eight acquisition scenarios. Every comparison with the atmospheric-only estimator uses the same variance, probes and error envelope. All 128 estimator/scenario combinations complete and produce 512 gas summaries. Their design objective is no worse than the atmospheric-only control within solver tolerance. We replay the 64 previously inspected atmospheric states, adding the same targetwise calibration worst case; this continuation does not call those states a new independent validation set.

VI. A RESOURCE-INDEPENDENT CALIBRATION FLOOR

Even if thermal noise vanishes, every estimator in this unbiased linear class has worst RMSE at least

$$\phi_j(\epsilon) = \min_{h^T D = e_j^T, h^T N = 0} \left\{ \max_s |h^T b_s| + \epsilon \|h\|_1 \right\}. \quad (10)$$

Split $h = h^+ - h^-$ with nonnegative parts and introduce $t \geq \pm b_s^T h$. This is a linear program with objective $t + \epsilon \mathbf{1}^T(h^+ + h^-)$. For its stored normalized equality matrix A_E , inequality matrix A_U , right-hand side d_E and positive cost c , any exact rational vectors satisfying

$$\mu \leq 0, \quad A_E^T \lambda + A_U^T \mu \leq c$$

give the lower bound $d_E^T \lambda$ by weak duality. The implementation clips inequality duals to the correct sign and scales both dual vectors until every inequality holds in exact rational arithmetic. The serialized coefficient arrays, rational vectors and a separate verifier make all 24 bounds machine checkable. Numerical primal/dual gaps are below 9.3×10^{-11} .

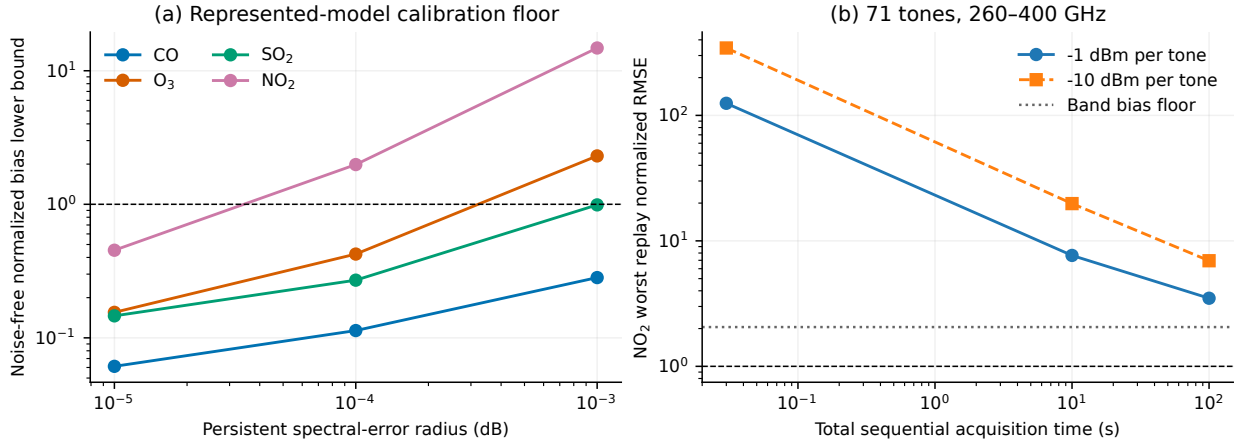


Fig. 1. Left: exact represented-LP dual lower bounds with all 202 probes. Right: NO₂ stress-replay RMSE at 10^{-4} dB persistent error, zero random residual and sequential band acquisition. The unit line is the declared magnitude-normalized error target. These are conditional calculations, not measured gas retrieval.

TABLE I
ROUNDED DISPLAYS OF CERTIFIED LOWER BOUNDS ON NOISE-FREE NORMALIZED WORST BIAS, ALL 202 PROBES. CERTIFICATES APPLY TO THE REPRESENTED NORMALIZED LP COEFFICIENTS.

ϵ [dB]	CO	O ₃	SO ₂	NO ₂
10^{-5}	0.061	0.155	0.146	0.453
10^{-4}	0.113	0.424	0.270	1.984
10^{-3}	0.283	2.303	0.990	14.806

At $\epsilon = 10^{-4}$ dB, the NO₂ lower bound is 1.9837335 with all probes and 2.0556340 in the 260–400 GHz band. Thus the declared four-gas error target cannot hold over this envelope for any estimator in the represented unbiased linear model, irrespective of pilot count. This is a conditional impossibility result for that model class and envelope. It is not a physical interval certificate for spectroscopy, a measured receiver limitation, or an impossibility theorem for nonlinear or prior-assisted retrieval. Bounds below one do not by themselves establish feasibility with noise.

VII. SEQUENTIAL ACQUISITION AND FINITE-RESOURCE RESULTS

A published VDI extender table lists a 260–400 GHz test-port band and typical output of -1 dBm; its dated March 2022 summary lists -10 dBm for that band [16], [17]. We retain both powers as scenarios. Typical magnitude stability of ± 0.5 dB in the current table is neither a measured spectral error for our setup nor the random residual SD in our likelihood. Dynamic range is also not used as an estimation precision specification.

The band contains 71 archived probes. With one tone at a time, pilot duration $T_s = 1 \mu\text{s}$, total duration T and inter-tone retuning time t_r , equal dwell permits

$$N_p = \left\lfloor \frac{T - 70t_r}{71T_s} \right\rfloor.$$

At zero retuning, 30 ms, 10 s and 100 s give 422, 140,845 and 1,408,450 pilots per tone. The 10 s, -1 dBm case

TABLE II
CALIBRATION-AWARE WORST RMSE ON THE 64-STATE STRESS REPLAY, $\epsilon = 10^{-4}$ dB AND ZERO INDEPENDENT RANDOM RESIDUAL. BAND ROWS USE SEQUENTIAL -1 dBm PROBING; FULL ROWS RETAIN IDEAL SIMULTANEOUS PROBING.

Acquisition	CO	O ₃	SO ₂	NO ₂
Full, 30 ms	0.290	1.422	0.809	12.594
Full, 10 s	0.140	0.802	0.447	2.425
Band, 10 s	0.276	1.328	0.658	7.654
Band, 100 s	0.230	0.957	0.515	3.489

radiates .0079433 J, compared with 1.9953 J for the legacy ideal simultaneous 23 dBm allocation. Thermal variance is rescaled with tone power. The original LEO path, apertures, noise figure and coherent likelihood remain assumptions; this source specification does not demonstrate a realizable satellite receiver. Retuning sensitivities of .1, 1 and 10 ms are archived, and infeasible short sweeps are retained.

In the legacy ten-million-pilot case, calibration-aware NO₂ improves from 3.385 for the atmospheric-only estimator to 2.425, yet fails the unit target. The earlier all-four success at zero persistent error is therefore a special point, not a robust operating specification. At 30,000 pilots CO and SO₂ retain replay worst errors below one (.290 and .809), while O₃ and NO₂ do not. Longer dwell in the single band helps its noise component but cannot remove the certified bias floor.

VIII. WHAT AN EXTERNAL VALIDATION MUST MEASURE

A bounded search of public spectroscopy and atmospheric measurement resources did not acquire synchronized 60–400 GHz attenuation and pollutant truth for this setup. The search log records both inspected alternatives and failed access attempts; it does not establish that such data do not exist. The next experiment must independently characterize frequency-dependent persistent error and covariance over the actual sweep, provide synchronized concentration or column truth and atmospheric profiles, and separate calibration from the final evaluation period. A useful protocol must fix the

concentration averaging period, acquisition power and time, and acceptance error before looking at final outcomes. No field-validation claim is made here.

IX. CONCLUSION

Joint atmospheric and calibration robustness improves the declared finite-resource objective, but the exact represented-model NO₂ bias floor rules out the four-gas target at 10⁻⁴ dB persistent spectral error within the unbiased linear class. Sequential band acquisition further exposes the cost of sharing observation time across tones. CO and SO₂ retain conditional margins in the tested envelope; extra pilots alone cannot repair persistent bias. The contribution is a reproducible robust-estimation and feasibility analysis, with machine-checkable algebra and explicit hardware assumptions. Measured calibration and synchronized external attenuation truth remain necessary.

REPRODUCIBILITY

The current 512 gas summaries, 32,768 state/gas replay errors, estimator operators, physical input hashes and acquisition scenarios are in `results/continuation_calibration`. The 24 serialized rational dual certificates and LP coefficients are in `results/continuation_calibration_floor`. A separate portfolio verifier checks every dual inequality in exact rational arithmetic and independently recomputes every stored risk and target/nuisance identity. Source-unit and download preflight failures are retained in the run logs. CPU software, wall time and sampled memory are recorded for each numerical run; all displayed tables and the current figure use the recorded CSVs. Earlier estimator checks and their rank-detection correction remain in `results/revision_0908`.

TABLE III

COHERENT MATCHED-MODEL ESTIMATION, ZERO RANDOM RESIDUAL. ERROR IS DIVIDED BY THE GAS REFERENCE MAGNITUDE. THE FINAL COLUMN BOUNDS ARBITRARY SYSTEMATIC ERROR PER PROBE.

N_p	Gas	Predicted	Simulated	ϵ_j (dB)
300	CO	0.994	0.981	0.000740
3,000	CO	0.314	0.310	0.006530
30,000	CO	0.099	0.098	0.006845
30,000	SO ₂	0.582	0.575	0.000703
30,000	O ₃	1.250	1.246	None
30,000	NO ₂	12.133	12.191	None

APPENDIX A

PRIOR ESTIMATOR AND ATMOSPHERIC CONTROLS

These controls retain the earlier resources and error models. In particular, zero persistent calibration error is a special case, superseded as a robustness specification by the current joint envelope. The 64 states called new in the earlier protocol are reused in the present stress replay.

A. Frozen computation and results

The grid contains two likelihoods, four pilot counts (30, 300, 3,000, 30,000), three independent marginal residual scales (0, 0.001, 0.01 dB) and two covariance shapes. For each configuration, 10,000 new Gaussian observation-noise draws validate the actual linear estimator against its predicted covariance. Common standard-normal draws pair the configurations; they are not independent replications of physical data. We also evaluate deterministic physical mismatch at componentwise training concentration medians and 95th percentiles. These quantile vectors are declared scenarios, not necessarily jointly observed atmospheric states.

Across all configurations the maximum empirical-to-predicted RMSE discrepancy is 1.57 percent. This is a simulation consistency check, not an uncertainty interval for receiver performance. Earlier independent-noise calculations allowed CO at 300 pilots only below 0.00490 dB random residual. Under zero random residual, the separate worst-case systematic allowance is only 0.000740 dB. These numbers answer different questions and cannot be substituted for one another.

The mismatch experiment holds the estimator and its nominal covariance fixed. The true forward mean uses $T + 5$ K, $p - 2$ kPa, gas height 1.2 km, or all three together, including their background change. Temperature and pressure are not supplied to the estimator as corrected values. Noise covariance remains the declared nominal one so that this experiment isolates mean-model error; it is not a complete altered-link simulation. The exact bias formula evaluates the result without Monte Carlo uncertainty.

Matched-model SO₂ falls below its reference error, but a 5 K temperature mismatch alone increases its ratio from 0.582 to 2.992; the combined mismatch gives 3.918. More pilots reduce noise but do not remove this persistent model bias. CO retains more margin at 30,000 pilots, while its 300-pilot result is fragile because its noise error already nearly reaches

TABLE IV

NORMALIZED RMSE AT 30,000 COHERENT PILOTS, ZERO RANDOM RESIDUAL AND MEDIAN CONCENTRATION SCENARIO. PERTURBATION AMPLITUDES ARE ASSUMPTIONS, NOT MEASURED RECEIVER ERRORS.

Actual forward model	CO	O ₃	SO ₂	NO ₂
Matched	0.099	1.250	0.582	12.133
$T + 5$ K	0.420	2.130	2.992	13.311
$p - 2$ kPa	0.102	1.503	1.162	12.434
1.2 km gas height	0.112	1.255	0.584	12.137
Combined	0.484	1.472	3.918	14.745

TABLE V

WORST NORMALIZED RMSE ACROSS THE 27-STATE GRID, 30,000 PILOTS AND MEDIAN CONCENTRATIONS. PARENTHESES GIVE MATCHED NOISE SD.

Gas	Nominal GLS	Atmospheric tangent GLS
CO	.556 (.099)	1.047 (1.046)
O ₃	3.190 (1.250)	2.653 (2.650)
SO ₂	4.487 (.582)	1.136 (1.134)
NO ₂	15.121 (12.133)	13.689 (13.684)

the target. Neither the chosen perturbations nor the worst-case spectral envelope constitute a calibrated uncertainty set for a real receiver.

B. Atmospheric nuisance projection and its cost

To address mean mismatch constructively, augment N with three tangent spectra of the complete modeled mean, including the background. The reference uses only the training median atmosphere and componentwise median concentrations. Central differences use temperature offsets ± 1 K, pressure offsets ± 500 Pa and gas-height offsets ± 100 m around 1.5 km. These derivative choices are frozen before evaluating the larger mismatch grid. Applying the same whitened SVD estimator with this expanded nuisance matrix removes those tangent directions exactly within numerical tolerance.

The evaluation uses all 27 combinations of T offsets $-5, 0, 5$ K, pressure offsets $-2, 0, 2$ kPa and gas heights 1.2, 1.5, 1.8 km. Both training median and 95th-percentile concentration vectors are retained. These are deterministic assumed perturbations, not independent physical observations. The forward mean is recomputed nonlinearly at every state; covariance remains nominal coherent noise with zero residual, at 3,000 or 30,000 pilots. Thus the grid tests curvature and composition mismatch beyond the local tangent fit without claiming a calibrated uncertainty set. SO₂ worst absolute normalized bias falls from 4.449 to .0724, but its noise SD nearly doubles. For CO the bias falls from .547 to .0462 while its noise SD increases more than tenfold. At 3,000 pilots the variance cost also makes tangent GLS worse for O₃ and NO₂; it is not uniformly beneficial. At the 95th-percentile concentrations and 30,000 pilots, tangent worst RMSEs are 1.061, 2.688, 1.184 and 13.865 in CO/O₃/SO₂/NO₂ order. None meets a normalized error of one over this grid.

Maximum numerical target-identity residual is 1.18×10^{-14} for the expanded estimator; all target ranks pass. Inputs, deterministic bias vectors, identities and complete grid re-

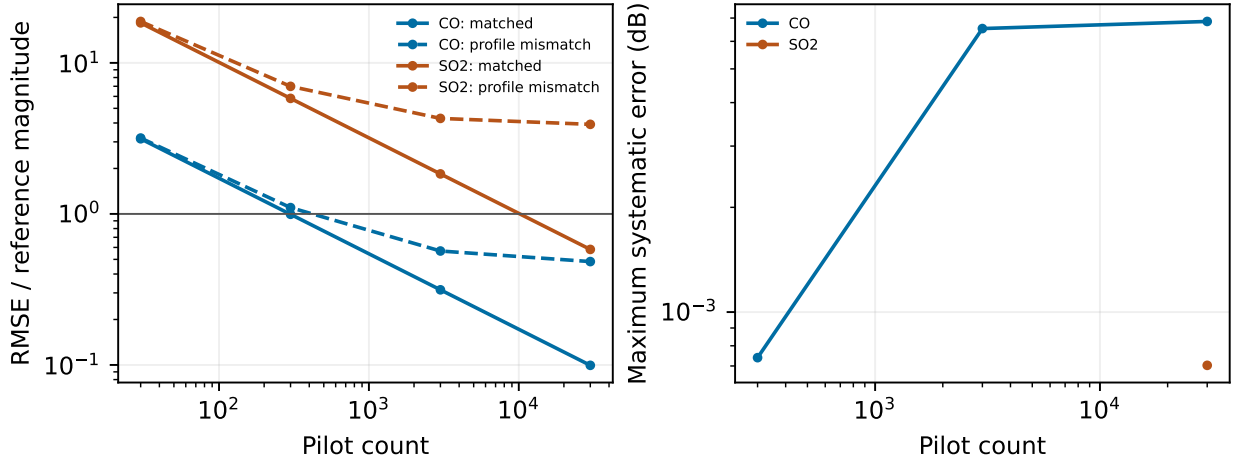


Fig. 2. Coherent estimation with zero random residual. Left: attainable matched-model error and combined physical mean mismatch at median concentrations. Right: exact targetwise allowance for arbitrary bounded systematic spectral error. Missing points already fail the noise target.

sults are in `results/closure_robust_atmosphere`. Independent recomputation checks the projected estimator and every bias/covariance expression. This repairs the missing robustness comparison while exposing, rather than concealing, its sensitivity/noise tradeoff. More measured atmospheric information or a different resource/design choice is needed to avoid freely discarding informative target directions.

C. Resource accounting and validation boundary

For a declared pilot duration T_s , the idealized simultaneous-probe observation duration is $N_p T_s$ and total transmitted energy is $P_{\text{total}} N_p T_s$. At the illustrative $T_s = 1 \mu\text{s}$ and 23 dBm, 300 pilots require 0.3 ms and 0.0599 mJ; 30,000 require 30 ms and 5.99 mJ. Retuning, coherent synchronization, guards and frequency-window access are excluded. The broad grid does not demonstrate that these probes can be acquired simultaneously with one front end. The larger budgets are therefore explicit resource scenarios, not free information gains.

A physical validation must supply paired attenuation and concentration/column truth, independent calibration recordings that estimate covariance and persistent bias across frequency and time, measured atmospheric profiles or justified uncertainty bounds, and a realizable frequency/power/timing allocation. The calibration data must be separated from the final evaluation period. A future acceptance criterion must specify the concentration averaging period and required estimation error before that evaluation. These are missing inputs, not completed experiments.

D. Balancing bounded atmospheric bias and variance

Hard tangent elimination treats each atmospheric direction as unbounded. This can discard useful gas information even though the declared perturbations are bounded. We instead retain target unbiasedness and the original nuisance cancellation while minimizing each target's worst MSE on a finite bias

envelope. Let $b_s, s = 1, \dots, 54$, be the archived nonlinear discrepancies for the 27 atmospheric states at the training median and 95th-percentile concentrations. These already inspected scenarios are design inputs, not a confirmation set. For target j solve

$$\begin{aligned} \min_{h \in \mathbb{R}^{202}, t \geq 0} \quad & h^T \Sigma h + t^2 \\ \text{s.t.} \quad & D^T h = e_j, \quad N^T h = 0, \\ & -t \leq b_s^T h \leq t \quad \text{for every } s. \end{aligned} \quad (11)$$

Here h is unrestricted in sign; only t is nonnegative. This is a convex quadratic/conic program. For the symmetric convex hull $\mathcal{B} = \text{conv}\{\pm b_s\}$, linearity and the triangle inequality give

$$\sup_{b \in \mathcal{B}} (h^T b)^2 = \max_s (h^T b_s)^2.$$

Thus the objective is the exact worst conditional MSE on the declared envelope. It is an application of convex robust estimation, not a claim that the envelope covers an actual atmosphere or receiver.

Writing $s_i = \sqrt{\Sigma_{ii}}$, we parameterize the whitened row as its minimum-variance solution plus a vector in the null space of the whitened target and nuisance constraints. Orthogonality makes the variance a constant plus a squared Euclidean norm. This avoids the initially unsuccessful dense solver parameterization. The implementation uses CLARABEL and a 10^{-10} relative cutoff for the projected bias-response rank; physical risks and the equality identities are recomputed from the resulting operator. These are numerical solutions, not interval certificates of optimality. Independent tests cover an analytic two-sensor bias/variance optimum, nuisance cancellation and the convex-hull envelope.

The bounded alternative reaches the error target for CO and SO₂ without increasing pilot count. Its design worst SO₂ error is .722, compared with 1.184 for hard tangent elimination when both concentration scenarios are included. CO improves to .218 instead of the hard method's 1.061. The earlier median-

TABLE VI

WORST NORMALIZED RMSE ACROSS THE 54 DESIGN DISCREPANCIES AT 30,000 COHERENT PILOTS, ZERO RANDOM RESIDUAL.

Gas	Nominal GLS	Hard tangent	Bounded bias
CO	0.707	1.061	0.218
O ₃	3.433	2.688	1.334
SO ₂	4.883	1.184	0.722
NO ₂	15.581	13.865	12.176

TABLE VII

WORST NORMALIZED RMSE ON ALL 64 NEW STATES, 30,000 PILOTS, ZERO RANDOM RESIDUAL. THE FINAL COLUMN COUNTS STATES MEETING ERROR ONE UNDER BOUNDED-BIAS ESTIMATION.

Gas	Nominal	Hard tangent	Bounded bias	Pass
CO	.601	1.049	.278	64/64
O ₃	2.785	2.706	1.312	0/64
SO ₂	3.294	1.182	.738	64/64
NO ₂	13.660	13.711	12.162	0/64

only table has a different denominator of scenarios and is not silently substituted here.

E. New nonlinear states and resource requirements

After fixing the bias envelope and solver, we draw 64 new states with seed 909510. Temperature and pressure offsets are uniform over $[-5, 5]$ K and $[-2000, 2000]$ Pa; gas height is uniform over $[1200, 1800]$ m. Each state uses a randomly selected complete real training-period concentration record, without filtering by outcome. The nonlinear forward mean is recomputed. These draws test new modeled states and compositions; the uniform perturbation law is not an empirically calibrated atmospheric distribution. No held-out outcome selects parameters or resources.

O₃ and NO₂ cannot meet the objective at these resources within the stated unbiased model: even the nominal matched thermal-only variance, retaining the original nuisances, requires at least 46,850 and 4,416,231 pilots after rounding upward. This is an information constraint, not a tuning failure. Geometric bisection of the bounded-bias design risk gives the following sufficient resources for its finite design envelope.

“Not reached” means the target still fails at 10^9 pilots; it is not an asserted universal impossibility. At the independently fixed 10^7 -pilot, zero-residual grid point, all four gases pass in all 64 new states, with worst errors .085, .519, .283 and .828 in CO/O₃/SO₂/NO₂ order. At one microsecond per pilot this requires ten seconds and 1.995 J of total transmitted energy under ideal simultaneous probing. The design-envelope NO₂ threshold alone requires about 5.91 seconds and 1.18 J. Neither resource statement demonstrates hardware feasibility, ten-second stationarity or a calibrated residual floor.

The complete design, held-out records, risk operators, failures, hashes and resource calculations are retained in `results/followup_bounded_bias`. The new estimator resolves the earlier CO/SO₂ variance penalty within the declared model. Synchronized measured attenuation, external

TABLE VIII

PILOTS NEEDED FOR NORMALIZED ERROR ONE ON THE FINITE DESIGN ENVELOPE. RANDOM RESIDUAL IS INDEPENDENT ACROSS PROBES; SYSTEMATIC CALIBRATION REMAINS SEPARATE.

Gas	Zero residual	.001 dB	.01 dB
CO	314	314	331
O ₃	58,973	67,768	Not reached
SO ₂	12,802	13,135	Not reached
NO ₂	5,909,703	Not reached	Not reached

calibration and a realizable receiver remain necessary for a field retrieval claim.

APPENDIX B

SUPPORTING PHYSICS AND EARLIER EXPERIMENTS

The following diagnostics document earlier questions and parameter conventions. Their full-period reference atmosphere, 256-probe low-SNR extrapolation and historical comparisons do not replace the training-only, 202-probe primary protocol above. Old results are retained to disclose their limitations and the evidence motivating the current question.

A. Related Work and Scope

Aliaga *et al.* integrated differential absorption radar with beyond 100 GHz satellite communications and evaluated communication and water vapour sensing tradeoffs using atmospheric models [1]. Their work supports the shared sensing and communication premise, but it neither uses the UCI pollutants nor demonstrates their retrieval. Yang *et al.* model molecular and particulate attenuation across space air ground media and compare gaseous attenuation with ITU recommendations [2]. Their decomposition motivates our layered gas and particle treatment, while also highlighting that particle composition and size cannot be replaced by an arbitrary spectral trend.

ITU-R P.676-13 describes gaseous attenuation and ITU-R P.835-7 supplies reference atmospheres [3], [4]. We use them as propagation context, not as a claim that our trace gas columns or exploratory PM parameters are ITU calibrated. HITRAN2024 was the current edition when the local line table was retrieved through HAPI in July 2026 [5]. HAPI supplies partition sums and a reference implementation check using the same spectroscopic records [6].

Dong *et al.* derive CRB and Bayesian CRB limits for Ku band rain sensing from satellite pilot attenuation [7]. Their pilot power variance model is methodologically relevant, but their rain coefficients and validation data are not transferred to sub THz pollutants. The 0.63 dB residual value used below is borrowed only as a scenario value. It is not calibrated for the present frequency range or hardware. The gap addressed here is thus an estimation limit study for multiple ambient pollutants using public surface records and line data, with simulated rather than measured link attenuation.

B. Real Data, Provenance, and Quality

1) *Evidence Boundary*: Table IX distinguishes external evidence from generated quantities. The Beijing data contain

TABLE IX
EVIDENCE AND ROLE IN THE STUDY

Item	Source	Count	Role
Spectra	HITRAN	9,340 lines	Real line positions, strengths, widths, shifts
Surface	UCI Beijing	383,585 rows	Real pollutant and meteorological scenarios
Column	ESA CCI	10 mo.	Real total CO and tropospheric NO ₂ retrievals
Lab THz	Mendeley	13 levels	Measured aqueous protein positive control
Health scale	WHO 2021	6 levels	Guideline comparison only
Sim. obs.	This study	20,000 rows	Simulated receiver observations

hourly measurements from 12 sites from March 2013 through February 2017 [8]. The ESA CCI records provide monthly total CO and tropospheric NO₂ columns at the same 39.5°N, 116.5°E grid cell from March through December 2013 [10], [11]. A CC BY Mendeley dataset supplies measured 0.9 to 1.3 THz aqueous protein spectra at 13 concentration levels across two proteins [9]. HITRAN parameters, UCI surface concentrations, satellite retrievals, and the laboratory spectra are real public data. The vertical shapes, LEO observations, receiver errors, and atmospheric estimator inputs are generated by the declared model. WHO values are health guideline concentration scales, not labels and not legal limits [12].

The processed HITRAN table spans 60 to 400 GHz and contains the main isotopologue for CO (15 lines), H₂O (37), O₂ (45), O₃ (1,301), NO₂ (1,536), and SO₂ (6,406). Its SHA256 is 7d063e4036da3d3e5... The cleaned UCI file SHA256 is 39d6ceee9d66824bcc... Full hashes, absolute input paths, package versions, and experiment assumptions are stored in the generated manifest.

2) *Air Quality Audit*: The original UCI record contains 420,768 station hour rows. Complete case processing retains 383,585 rows, or 91.163%. The retained table has no exact duplicate rows, no duplicate timestamp and station keys, no nonpositive target values, and no dew point above temperature. Station retention ranges from 86.704% to 94.616%.

The important failures are analytical rather than syntactic. In 17,642 rows (4.599%), PM₁₀ is below PM_{2.5}, so a physically nonnegative coarse mass cannot be formed. These rows are excluded from the estimator benchmark rather than silently clipped. Moreover, the real PM_{2.5}–PM₁₀ correlation is 0.8846, CO–PM_{2.5} is 0.7922, and NO₂–PM_{2.5} is 0.6707. A supervised model can therefore exploit population covariance even when a PM spectral component is undetectable. This is why regression accuracy alone is not accepted as evidence of direct spectral identification.

The complete case concentration fifth percentile, median, and ninety fifth percentile are respectively 200, 900, and 3,500 $\mu\text{g m}^{-3}$ for CO; 2, 45, and 177 $\mu\text{g m}^{-3}$ for O₃; 2, 7, and 59 $\mu\text{g m}^{-3}$ for SO₂; 8, 43, and 117 $\mu\text{g m}^{-3}$ for NO₂; 6, 55, and 242 $\mu\text{g m}^{-3}$ for PM_{2.5}; and 11, 82, and 279 $\mu\text{g m}^{-3}$ for PM₁₀. These records define realistic surface scenarios, not measured atmospheric columns.

3) *Satellite Column Audit*: The column experiment uses ten monthly files for each gas from March through December 2013. All 20 official objects downloaded anonymously, passed the declared NetCDF coordinate and variable checks, and retained valid values at the Beijing cell. CO source flags are 3 in every month and NO₂ quality flags are 1. The CO median is 3.317×10^{18} molecule cm^{-2} , its sample fifth to ninety fifth quantile span is 1.361×10^{18} , and the median reported uncertainty is 1.411×10^{17} . For tropospheric NO₂, the corresponding values are 1.304×10^{16} , 1.863×10^{16} , and 1.657×10^{15} molecule cm^{-2} . With only ten samples, the spans describe these available months rather than a stable population distribution. The products are coarse satellite retrievals with instrument and prior dependence; they are closer to path quantities than station concentrations, but they are not direct atmospheric truth.

C. Layered Atmospheric Forward Model

1) *Line by Line Gas Absorption*: Let $\tilde{\nu}_k$ denote wavenumber and let $T_0 = 296$ K. The HITRAN reference intensity $S_\ell(T_0)$ is adjusted in layer j as

$$S_\ell(T_j) = S_\ell(T_0) \frac{Q(T_0)}{Q(T_j)} \exp \left[-c_2 E_\ell'' \left(\frac{1}{T_j} - \frac{1}{T_0} \right) \right] \times \frac{1 - \exp(-c_2 \tilde{\nu}_\ell / T_j)}{1 - \exp(-c_2 \tilde{\nu}_\ell / T_0)}, \quad (12)$$

where $Q(T)$ is the HAPI partition sum and c_2 is the second radiation constant. Air pressure shift, air broadened Lorentz half width, and Doppler standard deviation are evaluated at each layer. The normalized Voigt function V_ℓ then gives

$$\sigma_{gjk} = \sum_{\ell \in g} S_\ell(T_j) V_\ell(\tilde{\nu}_k; T_j, p_j) \quad [\text{cm}^2 \text{ molecule}^{-1}]. \quad (13)$$

The lower atmosphere is divided into 24 equal thickness layers from 0 to 12 km. Temperature follows a standard lapse profile and pressure follows the corresponding barometric relation. Surface values are the medians of all retained UCI records: $T_s = 287.55$ K, $p_s = 101.04$ kPa, and dew point 2.9°C. This full period atmosphere is a declared transductive reference scenario, not a training only estimate. For target gas g , a measured surface mass concentration c_g is mapped to

$$n_g(z) = c_g \frac{N_A 10^{-12}}{M_g} \exp(-z/H_g) \quad [\text{molecule cm}^{-3}], \quad (14)$$

where c_g is in $\mu\text{g m}^{-3}$, M_g is in g mol^{-1} , and the assumed pollutant scale height is $H_g = 1.5$ km. The zenith attenuation design coefficient per unit surface concentration is

$$d_{kg}^{\text{zen}} = \frac{10}{\ln 10} \sum_j \sigma_{gjk} \frac{N_A 10^{-12}}{M_g} \exp(-z_j/H_g) \Delta z_j. \quad (15)$$

Here Δz_j is expressed in centimetres. H₂O number density is obtained from dew point and assigned a 2 km scale height;

O₂ uses a dry air mole fraction of 0.20946. Their integrated attenuation forms the background vector $\mathbf{b}$. Target gas and PM scale heights remain assumptions, not measured UCI profiles.

For an elevation e , the reference calculation applies the plane parallel factor

$$d_{kg}(e) = \frac{d_{kg}^{\text{zen}}}{\sin e}, \quad b_k(e) = \frac{b_k^{\text{zen}}}{\sin e}. \quad (16)$$

This is adequate for a controlled sensitivity study but not an Earth curved ray tracing claim.

At the UCI median surface state, direct comparison with HAPI 1.3.0.0 gives peak cross section relative errors from 4.44×10^{-8} to 6.38×10^{-7} across the six molecules. Normalized RMSE over the active grid is below 1.58×10^{-7} for every molecule. This validates the implemented Voigt arithmetic against HAPI; it does not validate the assumed atmospheric profiles against measured columns.

2) *Native Column Design*: The satellite experiment does not convert columns into surface concentrations. For target g , let N_g^{col} be the retrieved column in molecule cm^{-2} and let nonnegative profile weights w_{gj} sum to one over the modeled layers. The zenith attenuation coefficient per unit column is

$$a_{kg} = \frac{10}{\ln 10} \sum_j \sigma_{gjk} w_{gj} \quad [\text{dB}(\text{molecule cm}^{-2})^{-1}], \quad (17)$$

so the zenith target contribution is $a_{kg} N_g^{\text{col}}$. At elevation e , the same plane parallel rule as (16) gives $a_{kg}(e) = a_{kg} / \sin e$. The implementation attaches an explicit domain to every column and rejects a total column supplied to a tropospheric parameter or the converse.

The column atmosphere uses 40 layers from 0 to 20 km at the March to December UCI medians $T_s = 290.85$ K, $p_s = 100.75$ kPa, and dew point 6.9°C . CO is a total column with an 8 km reference scale height, while NO₂ is tropospheric with a 1.5 km scale height. The model forces the reported total CO column into a normalized 0 to 20 km profile and therefore omits its true upper atmospheric distribution. The resulting line shape bias has unknown sign; CO scale heights of 4, 8, and 12 km test it only partially. Floors are compared with the real monthly column span and reported retrieval uncertainty, never with a surface WHO concentration scale.

3) *Exploratory PM Extinction*: For a spherical particle of radius r , complex refractive index m , wave number k_0 , and density ρ_p , the Rayleigh approximation uses [13]

$$\Phi = \frac{m^2 - 1}{m^2 + 2}, \quad (18)$$

$$C_{\text{abs}} = 4\pi k_0 r^3 |\text{Im } \Phi|, \quad C_{\text{sca}} = \frac{8\pi}{3} k_0^4 r^6 |\Phi|^2, \quad (19)$$

$$\kappa_{\text{ext}} = \frac{C_{\text{abs}} + C_{\text{sca}}}{4\pi r^3 \rho_p / 3}. \quad (20)$$

The fine mode assumes diameter $1 \mu\text{m}$, density $1,500 \text{ kg m}^{-3}$, and $m = 1.50 + 0.01i$. The coarse mode uses $6 \mu\text{m}$, $1,800 \text{ kg m}^{-3}$, and $m = 1.53 + 0.01i$. Both remain within

TABLE X
REFERENCE LINK AND OBSERVATION ASSUMPTIONS

Parameter	Value	Parameter	Value
Altitude	550 km	Elevation	45°
Slant range	749.1 km	Total P_t	23 dBm
Tx aperture	0.5 m	Rx aperture	0.3 m
Aperture efficiency	0.65	Active probes	256
Bandwidth/probe	1 MHz	Noise figure	6 dB
Receiver T	290 K	Other loss	5 dB
Pilot symbols	30	Residual std.	0.63 dB

the configured Rayleigh size parameter limit of 0.3 over the grid. Mass decays with an assumed 1 km scale height. In the benchmark, fine mass is PM_{2.5} and coarse mass is $\max(\text{PM}_{10} - \text{PM}_{2.5}, 0)$. For a one parameter PM₁₀ bound, the fine fraction is fixed to the median valid UCI ratio, 0.7681. These particle properties are exploratory, not calibrated Beijing aerosol optics.

D. LEO Link and Pilot Observation

1) *Reference Link*: The conceptual probe grid contains 256 equally spaced frequencies from 60 to 400 GHz, a spacing of 1.333 GHz. It samples possible multiband sensing locations; it is not a proposed 340 GHz wide waveform. The received power on probe k is

$$P_{r,k} = P_{t,k} + G_t(f_k) + G_r(f_k) - L_{\text{FS},k} - b_k(e) - L_{\text{impl}}, \quad (21)$$

where the total transmit power is divided equally across active probes, aperture gain is $\eta(\pi D/\lambda)^2$, and thermal noise is kTB plus receiver noise figure. Table X records the reference assumptions.

At 45° , median background limited SNR is 14.97 dB, the fifth to ninety fifth percentile interval is $[-70.80, 20.64]$ dB, and 206 of 256 probes exceed 5 dB. Very opaque probes receive negligible Fisher weight.

2) *Pilot Attenuation Likelihood*: Following the pilot power argument in [7], the background referenced attenuation observation is

$$\hat{A}_k = \mathbf{d}_k^T \boldsymbol{\theta} + \beta_0 + \beta_b b_k + n_k, \quad (22)$$

where $\boldsymbol{\theta}$ is the target parameter vector, either surface concentration or native column amount, β_0 is an unknown offset, and β_b permits background amplitude mismatch. The independent Gaussian approximation uses

$$\text{var}(n_k) = \frac{c_0}{N_p} \left(1 + \frac{1}{\gamma_k} \right)^2 + \sigma_r^2, \quad c_0 = \left(\frac{10}{\ln 10} \right)^2. \quad (23)$$

Here γ_k is linear SNR, N_p is pilot count, $\sigma_k^2 = \text{var}(n_k)$, and $\sigma_r = 0.63$ dB is an assumed independent residual per probe. Independence is a declared scenario: real gain and atmospheric calibration errors can be correlated across frequency. At fixed marginal variance, correlation has no universal direction of effect after nuisance elimination, as the new covariance sensitivity demonstrates. Equation (23) is derived for moderate to

high SNR. We extrapolate it only to assign very large variance, hence nearly zero weight, to opaque probes.

As an observation model sensitivity, a deterministic coherently averaged complex channel gives the high SNR delta method term $2c_0/(N_p\gamma_k)$ before adding the same residual variance. Unlike (23), it has no pilot power high SNR floor. Neither likelihood has been calibrated against a sub THz receiver; both remain declared observation scenarios.

E. Fisher Bounds and Evaluation Protocol

1) *Nuisance Aware CRB*: For the surface gas result, let $\mathbf{D}$ contain the four gas design columns and $\mathbf{N}$ contain normalized nuisance columns for offset, $\text{H}_2\text{O}/\text{O}_2$ background, fine PM, and coarse PM. With $\mathbf{X} = [\mathbf{D} \ \mathbf{N}]$ and $\mathbf{W} = \text{diag}(1/\sigma_k^2)$, the Fisher information is [14]

$$\mathbf{J} = \mathbf{X}^T \mathbf{W} \mathbf{X}. \quad (24)$$

Eliminating nuisance parameters gives efficient gas information

$$\begin{aligned} \mathbf{J}_{\text{eff}} &= \mathbf{J}_{gg} - \mathbf{J}_{gn} \mathbf{J}_{nn}^{-1} \mathbf{J}_{ng}, \\ \text{Cov}(\hat{\boldsymbol{\theta}}) &\succeq \mathbf{J}_{\text{eff}}^{-1}. \end{aligned} \quad (25)$$

The reference gas block has rank four and condition number 775.4. We define the one sigma floor for target g as $\sqrt{[\mathbf{J}_{\text{eff}}^{-1}]_{gg}}$ and also report three times this value. These are local unbiased estimator bounds conditional on all declared physics.

Joint PM retrieval is not reported as identifiable. The fine and coarse design columns have correlation 0.999999995, and the smallest to largest singular value ratio of the concentration scaled six column design is 2.61×10^{-7} . Instead, we show two optimistic one dimensional bounds: fine PM alone and fixed composition total PM_{10} . These are best case references, not evidence that both PM fractions can be recovered.

2) *Probe Selection and Power Allocation*: For a selected set $\mathcal{S}$, the regularized design objective is [15]

$$\Phi(\mathcal{S}) = \log \det \left(\lambda \mathbf{I} + \sum_{k \in \mathcal{S}} \tilde{\mathbf{x}}_k \tilde{\mathbf{x}}_k^T \right), \quad (26)$$

where $\tilde{\mathbf{x}}_k$ is the transposed design row after variance whitening, weighted column normalization, and a common numerical rescaling. Greedy selection maximizes the incremental value of (26). The ridge is 10^{-9} times the mean candidate information per parameter. Column normalization prevents units alone from determining the selection; the objective is not WHO weighted and does not first project out nuisance directions. For each requested tone count, σ_k^2 is computed in advance using equal division of the final total power. The surface experiment selects from 1,024 candidates and the native column experiment from 512 candidates across 60 to 400 GHz.

Uniform and D optimal designs use the same total transmit power, divided equally among the active probes, and the link variance is recomputed for every tone count. For 32 selected probes, a separate guideline scaled allocator maximizes

$\log \det(\mathbf{J}_{\text{eff}})$ under the exact variance model. It scales the four gas parameters by their WHO concentrations, constrains power fractions to sum to one, and gives each probe at least 10^{-3} of its equal power fraction. This separates frequency placement from power redistribution. Increasing the nominal probe count also increases occupied bandwidth and the number of assumed independent residual samples in this experiment; it is therefore a design comparison, not a hardware scaling law.

3) *H_2O Local Control*: The H_2O control parameterizes an assumed 2 km water column by surface dew point. A centered finite difference with endpoints $\pm 0.25^\circ\text{C}$ around the training only median produces the local attenuation derivative on 512 candidate frequencies. The default D optimal design contains this derivative, an offset, four pollutant gas columns, and two PM columns. The strict design also contains the $\text{H}_2\text{O}/\text{O}_2$ background scale. Both use the same fixed total power and variance rule as the pollutant experiment. Bounds are evaluated for target only, offset plus pollutant nuisance, and the same set plus background scale. Endpoint forward evaluations check local nonlinearity, while paired training states at low and high humidity test transfer away from the reference temperature and pressure.

4) *Chronological Empirical Protocol*: After excluding the 17,642 PM ordering failures, 20,000 rows are selected deterministically over the sorted station hour table. A single median atmosphere generates attenuation from the real concentration vector, and Gaussian noise follows (23). Complete timestamp groups form 60% training, 20% validation, and 20% final test periods: 12,000, 4,000, and 4,000 rows. Training ends at 2015-09-10 01:00 and validation at 2016-06-03 04:00. The same 12 stations remain represented, so this tests temporal transfer but not unseen station or city transfer.

The training period mean is the data baseline. Ridge standardizes observations using training data only; α is selected only by validation normalized RMSE from 10^{-2} through 10^8 ; and the test set is evaluated once in the original benchmark. Per target RMSE is normalized by its training fifth to ninety fifth percentile range. Later test comparisons are diagnostic because follow up model exploration reused this period. Physical WLS uses the attenuation design directly without a label prior. An oracle joint WLS uses all gas and PM columns. A separate gas only test removes the true simulated PM contribution before inversion, then introduces a deliberate inverse mismatch of +5 K, -20 hPa, and a 1.2 km rather than 1.5 km pollutant scale height.

A post hoc metric audit fixes these rows, periods, training denominators, and targets. It adds a training prior LMMSE reference and a paired row bootstrap. The LMMSE uses only the training label mean and covariance as a Gaussian prior, then computes the posterior mean under the exact forward design and diagonal modeled noise covariance. It is therefore prior assisted and distinct from both supervised Ridge and label free physical WLS. Spectral model stability is tested over ten receiver noise seeds while holding real labels, forward signals, geometry, and the split fixed. Multitask ElasticNet and targetwise histogram gradient boosting configurations are

TABLE XI
REFERENCE FLOORS AT 45°; PEAK DB USES UCI Q95

Target	Peak dB	1σ $\mu\text{g m}^{-3}$	WHO $\mu\text{g m}^{-3}$	$1\sigma/\text{WHO}$	$3\sigma/\text{WHO}$
CO	0.03186	83,015	4,000	20.8	62.3
O ₃	0.00484	28,099	100	281.0	843.0
SO ₂	0.01517	4,308	40	107.7	323.1
NO ₂	0.00113	64,427	25	2,577.1	7,731.2
PM _{2.5}	0.00012	1,513,315	15	100,888	302,663
PM ₁₀	0.00014	1,581,204	45	35,138	105,414

frozen before this stability check.

The historical advanced diagnostic freezes 216 combinations of training or context prior, likelihood, probe objective, tone count, power allocation, nuisance projection, and non-negative posterior estimation. Validation alone selects every reported case. A second causal task uses 189 calendar, weather, station, and pollutant features whose target offsets are at least one hour. Station network reconstruction masks all six current query pollutant values before donor aggregation. Single channel repair instead masks only target j and permits the other five current query channels. Sentinel mutation tests enforce both masking contracts. These are ground sensor tasks and are not counted as THz inversion. The measured THz control fixes the 0.9 to 1.3 THz band without labels, averages five frequencies, and evaluates ordinary least squares by leaving one measured concentration level out; its metric is not directly comparable with the Beijing chronological split.

F. Results

1) *Signal Magnitude*: Fig. 3 shows excess attenuation at the UCI ninety fifth percentile concentration and 45° elevation. The strongest gas peak is only 0.03186 dB for CO. O₃, SO₂, and NO₂ peak at 0.00484, 0.01517, and 0.00113 dB. Exploratory PM peaks are approximately 1.2×10^{-4} dB. Every value is far below the assumed 0.63 dB residual component, while total pilot error is larger still because of the first term in (23).

The most informative individual probes agree with the line structure: CO information is concentrated near 345.33 GHz, O₃ near 358.67 GHz, SO₂ near 357.33 GHz, and NO₂ near 396 GHz, with additional NO₂ information in lower loss windows. Strong cross section alone is not sufficient because H₂O/O₂ background can also suppress SNR.

2) *Reference Detection Floors*: Table XI gives the main result. All reference one sigma bounds exceed the corresponding WHO concentration scales. The closest gas is CO at 20.8 times; NO₂ is 2,577 times. Three sigma ratios are exactly threefold larger. The PM values are already optimistic one mode bounds; joint fine and coarse retrieval is practically nonidentifiable.

The comparison is intentionally only a scale test. WHO values are health guidelines averaged over 8 h for O₃ and 24 h for the other targets, while the CRB describes one simulated attenuation burst and an assumed vertical mapping. The ratios do not establish a regulatory limit of detection, temporal averaging performance, or compliance classification.

TABLE XII
GAS PROBE FLOORS DIVIDED BY WHO SCALE [12]. REF. VALUES USE 256 UNIFORM OR OPTIMIZED PROBES FROM THE 1,024 POINT CANDIDATE GRID; OPTIM. USES THE COMBINED STRESS TEST.

Target	Ref. unif.	Ref. opt.	Optim. opt. 1σ	Optim. opt. 3σ
CO	20.85	11.55	0.353	1.059
O ₃	280.85	184.67	5.925	17.776
SO ₂	107.64	63.94	2.594	7.783
NO ₂	2,579.94	1,664.56	55.676	167.028

3) *Sensitivity and Optimistic Stress Test*: The elevation sweep has a sensing optimum at the lowest tested elevation. At 15°, the longer assumed pollutant path outweighs the lower link SNR, reducing one sigma gas ratios to 10.4 for CO, 135 for O₃, 71.9 for SO₂, and 1,294 for NO₂. The result is conditional on the plane parallel profile; it is not a general claim that the lowest elevation is operationally optimal.

Increasing pilots from 30 to 3,000 at 45° reduces the CO ratio from 20.8 to 12.8, after which the 0.63 dB residual dominates. Increasing total power from 23 to 33 dBm changes the CO ratio only to 20.4. Setting the residual to zero while retaining 30 pilots gives 16.4. These one factor results show that the gap is not simply a low SNR problem: weak and overlapping sensitivities, opaque frequencies, pilot power estimation noise, and nuisance projection all matter.

We also combine every optimistic setting evaluated: 15°, 3,000 pilots, zero residual, 33 dBm total power, and all 256 probes. This is a stress test, not a proposed deployment. Its one sigma floor to guideline ratios are 0.649 for CO, 9.19 for O₃, 4.35 for SO₂, 88.2 for NO₂, 3,538 for PM_{2.5}, and 1,232 for PM₁₀. Only CO crosses the guideline concentration scale at one sigma, and its three sigma ratio is 1.95. The combined case therefore does not provide a robust multi pollutant detection success.

4) *Optimized Probes and H₂O Control*: Table XII applies greedy D optimal selection to the denser 1,024 point candidate grid. At 256 probes, the reference gas ratios improve by factors of 1.52 to 1.81 over the matched uniform selection, yet none approaches one. In the combined optimistic case, optimized CO reaches 0.353 at one sigma but remains 1.059 at three sigma. The other three gases remain above their guideline scales even at one sigma.

Numerical power optimization under the stated variance model on the 32 selected probes adds only 0.2% to 4.3% improvement over equal power D optimal selection. Changing the regularizer from 10^{-6} to 10^{-12} leaves the 256 selected frequencies and floors unchanged. For optimistic CO, a one sigma ratio of one would require the independent residual to be at most 0.223 dB; a three sigma ratio of one is unattainable in this scenario even at zero residual. Reference case gases remain above one even at zero residual.

H₂O provides a modeled strong species control. Around the training only median state of 4.0°C dew point, 16.0°C temperature, and 1008.7 hPa, the 256 probe default design gives a local dew point floor of 0.290°C with offset, pollutant gas, and PM nuisance columns. When a background scale

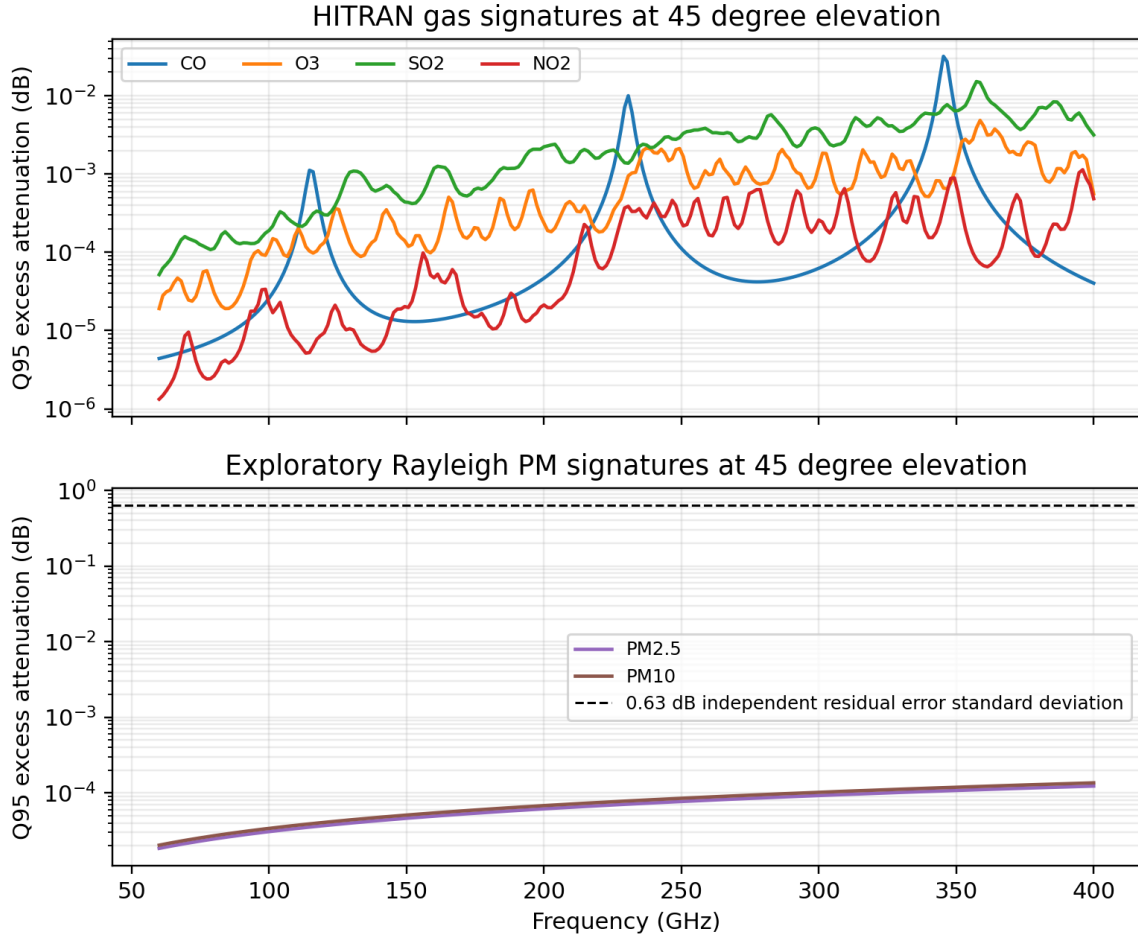


Fig. 3. Simulated excess attenuation for the complete case UCI ninety fifth percentile concentrations at 45° . Gas curves use the retrieved HITRAN line strengths. PM curves use the exploratory particle assumptions. The dashed line is only the assumed independent 0.63 dB residual component, not the total pilot error and not a measured instrument specification.

nuisance is added after that selection, the floor degrades to 2.541°C . A strict design that includes this nuisance during selection reduces the floor to 0.848°C , or 2.543°C at three sigma. Across pseudoinverse cutoffs from 10^{-10} to 10^{-15} the strict floor changes by less than 0.2%. Fixed temperature and pressure nonlinear checks remain within 8.2% relative error at the strict three sigma scale. Transfer to paired low and high humidity meteorological states produces 35% to 97% relative error, so this is a local information control rather than a global dew point estimator or measured retrieval.

5) *Real Satellite Column Stress Test*: Across the ten available months, the CO sample fifth to ninety fifth quantile span produces only 0.00597 dB peak zenith attenuation, while the tropospheric NO_2 span produces 6.54×10^{-5} dB. Table XIII compares optimized local floors with each observed span Δq and the median retrieval uncertainty u . Even without nuisance parameters, the reference floor ratios are 76.96 for CO and 2,772.99 for NO_2 . Declared nuisance columns increase them to 82.89 and 5,476.27. The combined optimistic scenario still gives one sigma ratios of 2.696 and 179.97, with three sigma

TABLE XIII
NATIVE COLUMN RATIOS WITH DECLARED NUISANCE [10], [11]. Δq IS THE TEN MONTH SAMPLE QUANTILE SPAN; u IS MEDIAN REPORTED UNCERTAINTY.

Target	Peak dB	Ref./ Δq	Optim./ Δq	Ref./ u
CO total	5.97×10^{-3}	82.89	2.696	799.7
NO_2 trop.	6.54×10^{-5}	5,476.27	179.97	61,559

ratios of 8.09 and 539.91.

Changing the CO scale height from 4 to 12 km changes its reference nuisance ratio from 107.1 to 74.7, which is material but does not change the conclusion. The native column result removes the surface concentration to column unit mismatch for these two species. It does not supply measured THz attenuation, and the ten monthly retrievals are too few and too coarse for an empirical sensing model.

6) *Empirical Estimation and Information Classes*: Validation originally selects Ridge $\alpha = 10^5$. On the chronological test period, its macro training Q05 to Q95 normalized RMSE

is 0.34744 versus 0.34748 for the training period mean; mean R^2 is -0.08293 versus -0.08316 . Thus the apparent learned estimate is numerically a mean predictor, not evidence that the simulated spectrum carries usable target information.

The audit reproduces the Ridge score to 2.22×10^{-16} . The paired bootstrap difference, Ridge minus mean, has a 95% interval of $[-1.50 \times 10^{-4}, 7.64 \times 10^{-5}]$, so no difference is resolved under this conditional row bootstrap. The training prior LMMSE reference is 0.347435. Across the ten fixed noise realizations, Ridge is 0.347476 ± 0.000011 , where the latter value is the sample standard deviation across modeled noise seeds, not a confidence interval. Multitask ElasticNet is 0.347741 and histogram gradient boosting is 0.347583. The frozen candidates beat Ridge on zero and one of ten seeds, respectively, so neither passes the predeclared eight seed stability rule. The alternative coherent CSI delta method sensitivity reaches only 0.345600 even after its independent residual is removed.

Previously explored forecasting, station reconstruction, single-channel repair and aqueous-protein controls are retained in the archived experiment notes. Their inputs and tasks differ from atmospheric THz inversion, so their scores are not used as sensing evidence here.

For the joint physical WLS, test RMSEs are $84,522 \mu\text{g m}^{-3}$ for CO, 27,845 for O₃, 4,223 for SO₂, and 58,383 for NO₂, close to the reference CRB scale in Table XI. Approximately half of unconstrained WLS predictions are negative, as expected when noise dominates a zero centred linear inversion. Joint PM WLS is numerically unstable, with RMSEs of 6.36×10^9 and $1.45 \times 10^9 \mu\text{g m}^{-3}$ for PM_{2.5} and PM₁₀. The optimistic one parameter PM bounds must not be read as joint retrieval performance.

In the gas only oracle test, changing temperature, pressure, and scale height in the inverse model increases each gas RMSE by 31.5% to 32.6%. This controlled mismatch confirms that even the already poor matched inversion is optimistic. A constrained estimator could remove negative outputs, but it cannot create absent Fisher information.

7) *Training-only and calendar-block review replay*: The historical atmosphere used medians from the entire observation period. The review recomputes temperature, pressure, dew point, and the declared PM composition summary using training rows only. A mutation test verifies that changing future weather cannot alter the training atmosphere. The same frozen likelihood and Ridge validation grid are retained; the new run evaluates the selected model on the existing test period, which has already been inspected in earlier work. This is repaired computation on reused data, not an untouched confirmation set.

For the uncertainty calculation, all selected station rows inside each complete calendar block are resampled together. Seven-day and thirty-day partitions test sensitivity to the assumed dependence horizon. The estimand remains the paired difference in macro training-range normalized RMSE. The block intervals account for within-block station and temporal dependence, but cannot account for arbitrary dependence

TABLE XIV
TRAINING-ATMOSPHERE ONE-SIGMA BOUNDS DIVIDED BY THE STATED WHO REFERENCE CONCENTRATION SCALES. RANGES COVER THE DECLARED COVARIANCE GRID ONLY.

Gas	Independent	Minimum	Maximum
CO	21.20	19.39	23.59
O3	289.41	279.95	352.09
SO2	111.17	103.92	125.72
NO2	2706.19	2489.12	2878.15

across blocks or variation between cities. The thirty-day analysis has only ten observed blocks and is particularly limited.

The training-only replay gives Ridge 0.34744429 and mean 0.34747886. The thirty-day paired difference interval is $[-1.0382 \times 10^{-4}, 3.2635 \times 10^{-5}]$; the seven-day interval also includes zero. The historical replay with calendar blocks likewise resolves no difference. The earlier row-bootstrap interval is retained above only as historical methodology and does not serve as the revised uncertainty claim. Raw predictions, test row indices, both block lengths, selected alpha, input hashes and training atmosphere are stored under `results/review`.

8) *Training-atmosphere correlated-noise sensitivity*: Independence is not a universal optimistic assumption when marginal noise variances are fixed. A correlated component aligned with an eliminated offset can be less damaging than independent errors of the same marginal variance, whereas a component aligned with a target spectral signature can be more damaging. For example, with two probes, target column $(1, -1)^T$, an unknown common offset, unit marginal variance and a common residual variance 0.4, the target variance bound is 0.3, compared with 0.5 for independent noise. A separate numerical test verifies this elementary counterexample.

The new physical sensitivity uses only the training-derived atmosphere of the repaired spectral replay: 290.35 K, 100,800 Pa and 5.4°C dew point. A check recomputes these quantities from training rows. The PM fine fraction is likewise training-derived. Let v_k denote the original total marginal variance and let $\sigma_r = 0.63$ dB remain the declared residual scenario. We keep every marginal variance unchanged and use

$$\Sigma = \text{diag}(v_k - \sigma_r^2) + \sigma_r^2 R, \quad R = (1 - r)I + r[e^{-|f_k - f_l|/\ell}]_{kl},$$

with $r \in \{0, 0.5, 0.9\}$ and $\ell \in \{1, 10, 100\}$ GHz. All nine covariance scenarios are frozen and retained. Whitening by the full covariance precedes nuisance elimination. We include an offset, background and both PM nuisance directions with a declared 10^{-12} relative SVD cutoff after column normalization; the nuisance rank is four in every case.

The fixed-marginal covariance changes can raise or lower the bound; for CO the range is 19.39–23.59 times the reference concentration scale, versus 21.20 under independence. No target approaches its reference scale in this grid. This reinforces the negative result only for the stated scenarios. It supplies neither a calibrated hardware covariance nor a worst-case envelope over all correlated errors. The reference atmosphere here differs from the historical full-period atmo-

TABLE XV
CONDITIONAL ONE-SIGMA REQUIREMENTS. ZERO-RESIDUAL RATIO AND
MAXIMUM INDEPENDENT RESIDUAL (dB). ALL FOUR GASES AT THE
LARGEST COHERENT PILOT COUNT ARE SHOWN.

Model	N_p	Gas	Ratio	Max. σ_r
Power	30	CO	16.894	None
Power	300	CO	5.342	None
Power	3000	CO	1.689	None
Power	30000	CO	0.534	0.04055
Coherent	30	CO	3.144	None
Coherent	300	CO	0.994	0.00490
Coherent	3000	CO	0.314	0.04511
Coherent	30000	CO	0.099	0.04769
Coherent	30000	O ₃	1.250	None
Coherent	30000	SO ₂	0.582	0.00662
Coherent	30000	NO ₂	12.133	None

sphere in the earlier physical tables; those tables remain a declared historical scenario, not a prospective training-only calculation. Full configuration, input hashes and bounds are in `results/repair_correlated`.

9) *Conditional Receiver Requirements*: The preceding negative grids do not rule out all receiver/noise/resource combinations. We therefore compute explicit requirements rather than extrapolate a single borrowed residual value. The training-only atmosphere and reference link are retained. Both likelihoods now use the same 202 probes whose modeled SNR is at least 5 dB, avoiding low-SNR extrapolation in this comparison. With $c = (10/\log 10)^2$, linear SNR γ_i , pilot count N_p and independent residual standard deviation σ_r , the declared alternatives are

$$v_i^{\text{power}} = \frac{c}{N_p} (1 + \gamma_i^{-1})^2 + \sigma_r^2, \quad (27)$$

$$v_i^{\text{coherent}} = \frac{2c}{N_p \gamma_i} + \sigma_r^2. \quad (28)$$

These are conditional likelihood scenarios, not measured receivers. Whitening and nuisance projection include an offset, background scale and both PM columns, with normalized nuisance SVD cutoff 10^{-12} . The four gas concentration columns are scaled by the same magnitude references as before. The resulting ratios are local one-sigma model bounds, not detection probabilities or compliance criteria.

The frozen grid uses 30, 300, 3,000 and 30,000 pilots and residual standard deviations 0, .001, .01, .1 and .63 dB. For each model and target, bisection reports the maximum independent residual in $[0, .63]$ dB for which the local floor is no larger than its reference scale. “None” means the target already fails at zero residual within that row’s resources.

Coherent CO crosses the one-sigma scale at 300 pilots only with a residual below .00490 dB; 3,000 pilots allow .04511 dB. At 30,000 pilots, coherent SO₂ also crosses, but requires residual below .00662 dB. O₃ and NO₂ still fail even at zero residual in the largest coherent row. Power-domain CO requires 30,000 pilots in this grid. Increasing pilot count increases observation time and total pilot energy at fixed per-pilot power; these rows are not equal-resource alternatives. The

boundary is conditional on independent residuals and does not override the earlier covariance sensitivity.

The results in `results/resolution_noise_requirements` replace a blanket negative reading with explicit receiver requirements. They do not supply measured synchronized atmospheric attenuation, validate the physical profiles or turn a local bound into successful pollutant retrieval.

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